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M506I Research Methods and Scientific Work (SS0325)

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Impact of Gamification on Learning Retention

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June 2025

I. Abstract

***Abstract*—Gamification has become a powerful tool in modern education, aiming to broaden the learners' motivation and engagement through game-based mechanics such as points, leaderboards, badges, and narrative challenges, etc. While several studies have demonstrated its short-term motivational benefits, its long-term impact on learning retention remains underexplored and often inconclusive. This paper explores the effect of gamification on long-term learning retention compared to traditional instructional methods. Drawing upon a systematic literature review and recent empirical findings including the work of García Iruela et al. (2022), and Çakıroğlu et al. (2021) this study examines how specific gamified elements contribute to learning retention. It further explores how perceived motivation and engagement mediate the effectiveness of gamification. The study adopts a conceptual framework that maps the relationship between gamified interventions, motivational dynamics, active learning processes, and retention outcomes. The findings are expected to inform educators, instructional designers, and e-learning developers on how to strategically design gamification that supports sustainable learning outcomes beyond initial engagement.**

II. Research Problem, Questions, and Objectives

A. Research Problem

Despite the growing interest and implementation of gamified learning environments, there remains a persistent challenge in ensuring robust and long-term learning retention across various educational settings. Many traditional teaching methods may not adequately address the cognitive and motivational factors critical for sustained knowledge recall. While gamification has shown promise in enhancing immediate engagement, the specific mechanisms through which it contributes to improved *learning retention* and the conditions under which this impact is maximized are not yet fully understood or consistently demonstrated in empirical research. This gap necessitates a focused inquiry into how gamified elements can be strategically designed and implemented to directly foster better knowledge retention.

B. Research Questions

1. To what extent does the integration of gamified elements in an educational setting influence students' long-term learning retention compared to traditional learning methods?
2. What is the the mediating role of entertainment and motivation in the relationship between gamification and students' long-term learning retention?
3. How does specific gamified elements(Badges, Rewards, Leaderboards, Challenges) contribute to the overall gamification's effectiveness in an educational setting?
4. How do students' perceptions of engagement and motivation in a gamified learning environment mediate or influence their learning retention?
5. How do technological constraints, repetitive content, and superficial engagement negatively influence the entertainment derived from gamified learning, thereby impacting motivation and learning retention?

C. Research Objectives

1. Investigate and quantify the impact of a gamified learning intervention on long-term retention of students' learning compared to the traditional method.
2. Analyze the mediating pathways through which enrollment and subsequent motivation influence student long-term learning retention in gamified settings.
3. Identify and analyze the specific gamified elements that students report as the most impactful on their retention of knowledge.
4. To quantify overall impact of a gamified learning intervention on students'long term learning retention compared to a non-gamified control.
5. Explore the inhibitory effects of technological constraints, repetitive content, and superficial engagement on learning entertainment, motivation, and ultimately retention within gaming contexts.

III. Hypothesis

- **Null Hypothesis (H_0):** There is no significant difference in long-term learning retention between students exposed to a gamified learning environment and those in a traditional, non-gamified learning environment.
- **Alternative Hypothesis**
  - $\mathbf{H_1}$: Badges will positively impact the overall gamification elements.
  - $\mathbf{H_2}$: Rewards will positively impact the overall gamification elements.
  - $\mathbf{H_3}$: Leaderborads will positively impact the overall gamification elements.
  - $\mathbf{H_4}$: Challenges will positively impact the overall gamification elements.
  - $\mathbf{H_5}$: Gamification elements will positively influence **Entertainment**.
  - $\mathbf{H_6}$: Technological constraints will negatively influence **Entertainment**.
  - $\mathbf{H_7}$: Repetitive Content will negatively influence **Entertainment**.
  - $\mathbf{H_8}$: Superficial Engagement will negatively influence **Entertainment**.
  - $\mathbf{H_9}$: **Entertainment**will positively influence**Learning Retention**.
  - $\mathbf{H_{10}}$: **Entertainment** will positively influence **Motivation**.
  - $\mathbf{H_{11}}$: **Motivation** will positively influence **Learning Retention**.
  - $\mathbf{H_{12}}$: The integration of gamified elements in an educational setting will lead to a significant improvement in students' long-term learning retention compared to traditional non-gamified learning methods.

IV. Conceptual Framework

The framework illustrates the hypothesized relationships between gamification elements, constraints, participation quality, and learning retention among students.

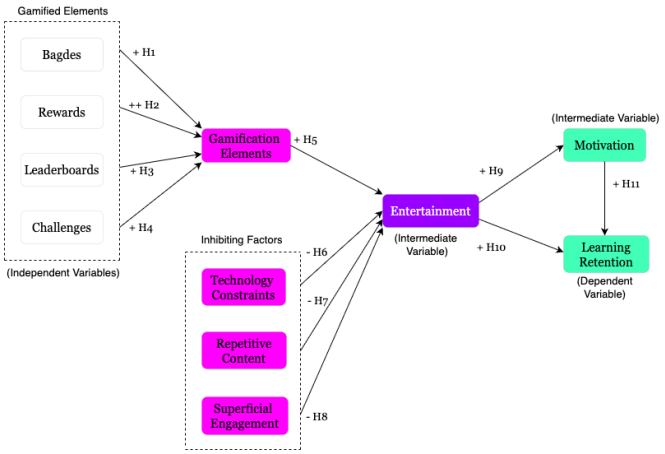


Fig. 1. Conceptual Framework for Gamification and Learning Retention

A. Explanation of the Model Architecture

- **Independent variables: Gamififcation Elements**(composed of Badges, Rewards, Leaderboards, and challenges) serve as the primary independent variables. Each element is hypothesized to positively contribute to the overall gamification construct(arrows labeled +H1, ++H2, +H3, +H4).
  - **Badges (+H1)**: Digital markers of achievement.
  - **Rewards (++H2)**: Tangible or intangible incentives.
  - **Leaderboard (+H3)**: A ranking system.
  - **Challenges (+H4)**: Tasks or quests to be overcome.
- **Inhibiting Factors: Constraints** (Technological constraints and Repetitive content) and **Participation**(Superficial Engagement) are modeled as moderating or inhibiting factors, hypothesized to negatively impact **Entertainment** (arrows labeled -H6, -H7, -H8).
  - **Technological Constraints (-H6)**: Limited access to devices or unstable network ccnnections can negatively impact the entertainment derived from gamification, reducing motivation or engagement when students cannot participate fully.
  - **Repetitive Content (-H7)**: Overly repetitive challenges or content can reduce the entertainment and engagement among users.
  - **Superficial Engagement (-H8)**: Over-relience on game mechanics without meaningful content can result in surface-level participation, with learners engaging only for rewards than fostering deep understanding, thereby negatively impacting entertainment and actual learning outcomes.
- **Intermediate Variables: Entertainment** and **Motivation** act as mediators in the pathway from gamification elements to learning retention. Gamification elements are expected to enhance **Entertainment**(+H5), which in turn boosts **Motivation**(+H10) and directly improves **Learning retention**(+H9). **Motivation** further positively influences **Learning Retention**(+H11).
- **Dependent Variable: Learning Retention** is the main dependent variable, representing students' ability to retain knowledge following gamified learning learning interventions.

This model visually summarizes the expected positive and negative influences among variables in the study, clarifying the roles of independent, mediating, and dependent variables in the context of gamification's impact on students' learning retention

Additionally, negative factors such as superficial engagement, repetitive or poorly designed gamified content, and technological constraints may moderate the effectiveness of gamification.

V. Literature Review

A. Review of Relevant Literature

Gamification, which originally arose from the industry of digital consumer products, appears to have found a welcoming home in education due the potential to tackle high levels of disengagement and poor levels of learning. Much like in other industries, there is considerable literature showing that gamification can enhance motivation and engagement Hamari et al. (2014), Seaborn & Fels (2015). Many of these studies have pointed to how elements like points, badges, and leaderboards can offer extrinsic motivations while challenges and stories can engage learners intrinisically by providing a sense of accomplishment and meaning Ryan & Deci (2000). Research by Kapp (2012) illustrates how gamification can turn passive learning into active iterative learning and, consequently, encourage learners to engage greater with content material.

There are now a few systematic reviews that show the benefits of gamification in education. For example, Subali et al. (2021) found in a systematic review focused on language learning that gamification had a positive effect on motivation, engagement, and learning performance and called attention to the positive effects of the elements inherent in gamification, for example, points, badges, leader-board and challenges (contrary to gamification arguments, not in favor of the players engagement the study didn't imply that). Similarly, Rachman & Saparudin (2024) found that gamification made English language learning more enjoyable and less stressful- ultimately leading to increased levels of active participation, especially related to vocabulary awareness/retention and grammar retention.

The significance of entertainment in education, commonly referred to as edutainment, is also gaining momentum. As suggested by Prensky2001, combining education and entertainment can elicit a higher level of engagement in learning. Gamification facilitates this naturally, as it blends fun and game-like possibilities to the more traditional learning environments. Studies show that when learners perceive tasks as enjoyable, they tend to persist more, learn more deeply, and process information more robustly which all contribute positively to remembering Malone & Lepper (1987).

While engagement is an important part of the learning process, we still need to remember that retention of knowledge is still the final goal of educational experiences. Across the literature, there is generally support for the positive relationship of gamification for retention. For instance, in one study, Anderson & Dill (2000), learning tasks which required practice and feedback which is central to gamification, were found to promote memory retention. However, when looking at the literature in totality, the results are not entirely positive. Dicheva2015 concluded that the effectiveness of gamification is often contingent on how gamification is constructed and implemented in the learning environment. Similarly, Sailer & Homner (2020) found in a meta-analysis of 51 studies that the effect of gamified learning on learning outcomes was moderate, although it was positively related to cognitive outcomes including retention of content. Of particular note, they found that certain aspects of gamified learning appeared particularly effective in educational environments, immediate feedback and structured rehearsal (ex. quiz, leveling up, etc); as these facilitate the processes of memory consolidation and application of active recall Hattie & Timperley (2007).

That said, gamification is not guaranteed to have an impact. There are a number of restrictions that can limit the impact of gamification. For example, Subali et al. (2021) noted that technological restrictions such as restricted internet access can impact participation and learning outcomes. Similarly, if the section of the game with the turning point has multiple innings, repetition of content can promote disengagement and lower the entertainment value. Increased points and badges as extrinsic motivation can also lead to superficial engagement, as Kohn (1999) suggested. Supporting that argument, López-Cobo et al. (2020) stated that gamification tends to help motivation, and that the influence on academic performance is very inconsistent. A poorly deployed gamified system can even have retribution if ineffective, as it can promote disengagement with the information, lowering retention if learners completely disengage or do not internalize the information.

In this regard, the study will position its analysis in the larger research debate; discussing to what extent gamification can enhance retention through entertaining and motivating design will improve learning gain and retention and addressing the factors that potentially limit its effectiveness.

B. Key Theories Related to Gamification

KEY CONCEPTS

Gamification: At its core, gamification involves applying game-design elements (e.g., points, badges, leaderboards, levels, challenges, narratives, avatars, virtual goods, instant feedback) and game-thinking (e.g., player progression, mastery, social connection) in non-game contexts, typically to drive engagement and motivate specific behaviors (Deterding et al. 2011*a*). The key is not to turn learning into a game, but to incorporate game-like mechanics to make learning more compelling.

Learning Retention: This refers to the ability to recall and apply previously learned information over time. It is distinct from immediate understanding or short-term memory. Factors influencing retention include the depth of processing during learning, the frequency of retrieval practice, the presence of meaningful connections, and the emotional context of learning.

Self-Determination Theory (SDT): Proposed by Deci & Ryan (1985*a*), SDT is a macro theory of human motivation and personality. It posits that individuals have three innate psychological needs:

- **Competence:** The need to feel effective and capable in one's actions. Gamification often provides clear progression, feedback, and challenges that satisfy this need.
- **Autonomy:** The need to feel volitional and self-initiating in one's behaviors. Choice in quests or personalized learning paths in gamified systems can foster autonomy.
- **Relatedness:** The need to feel connected to others and to belong. Social features like leaderboards, team challenges, or collaborative quests in gamified environments address this need.

When these needs are met, individuals are more likely to experience intrinsic motivation, leading to deeper engagement and better learning outcomes, including retention.

Flow Theory: Developed by Csikszentmihalyi (1990*a*), Flow Theory describes a mental state in which a person performing an activity is fully immersed in a feeling of energized focus, full involvement, and enjoyment in the process of the activity. This state is highly relevant to *entertainment* in gamification. When the challenge of a task matches a person's skills, they are more likely to enter a flow state, which can positively impact learning and retention.

Cognitive Load Theory (CLT): Developed by Sweller (1988), CLT focuses on how information is processed and stored in working memory. While poorly designed gamification could increase extraneous cognitive load, well-designed gamified elements (e.g., through clear progression, immediate feedback, and structured challenges) can reduce extraneous load and optimize germane load, aiding memory consolidation and retention.

C. Theoretical Framework for Answering Your Questions

The theoretical framework for this research is primarily rooted in **Self-Determination Theory (SDT)** and **Flow Theory**, supplemented by insights from **Cognitive Load Theory (CLT)** and principles of active learning derived from constructivist approaches.

SDT provides the core lens through which we understand how gamification enhances *motivation* and *engagement* Deci & Ryan (1985*b*). By designing gamified learning experiences that strategically fulfill students' psychological needs for competence (e.g., through progress bars, levels, achievements), autonomy (e.g., choice in activities, personalized paths), and

relatedness (e.g., leaderboards, team challenges), intrinsic motivation is hypothesized to increase. This heightened intrinsic motivation leads to more sustained effort, deeper cognitive engagement, and a more positive attitude towards learning.

Flow Theory explains the mediating role of **entertainment** Csikszentmihalyi (1990*b*). When gamified learning is well-designed, it creates a balance between challenges and skills, leading to a state of flow where learners are deeply immersed and find the experience inherently enjoyable. This entertainment factor is hypothesized to directly enhance learning retention by promoting deeper processing and more positive emotional associations with the content. Furthermore, entertainment derived from the learning process is expected to significantly boost **motivation**, which in turn further strengthens the pathway to improved retention.

This increased engagement and motivation, in turn, facilitate **active learning**. Gamified elements often necessitate active participation, problem-solving, and repeated interaction with the content, moving beyond passive reception of information. This active engagement, combined with frequent and immediate feedback (a common gamification feature), aligns with cognitive science principles that suggest active retrieval practice and elaborative rehearsal are critical for solidifying memories and enhancing *long-term retention*.

Furthermore, **Cognitive Load Theory** informs the design considerations and the impact of inhibiting factors Sweller (1988). Effective gamification should reduce extraneous cognitive load by making learning goals clear and feedback immediate, allowing learners to focus their cognitive resources on germane load (the mental effort for schema formation). However, **technological constraints** (e.g., technical glitches, poor internet) and **repetitive content** can increase extraneous load and disrupt the flow state, thereby negatively impacting entertainment, motivation, and retention. Similarly, an overemphasis on extrinsic rewards can lead to **superficial engagement**, where learners bypass deeper processing in favor of quickly earning rewards, thus undermining the long-term retention goals.

Therefore, the theoretical framework posits that gamification, by leveraging game design elements Deterding et al. (2011*b*) to enhance intrinsic motivation (SDT) and foster an entertaining "flow" state (Flow Theory), promotes active learning and optimizes cognitive load (CLT), ultimately leading to more effective encoding, consolidation, and retrieval of information, thereby improving learning retention. The research questions will be answered by exploring these linkages empirically, measuring retention outcomes, identifying influential gamified elements, assessing the mediating role of student motivation and entertainment, and understanding the inhibitory effects of constraints and superficial engagement.

VI. Methodology and Method

A. Research Methodology

This study adopts a **quantitative method**, which is appropriate for investigating measurable impacts of gamification on learning retention. Quantitative analysis allows for objective comparison of numerical outcomes (e.g., test scores, reaction times, accuracy rates) across multiple datasets. The secondary nature of this methodology ensures that the research remains feasible within time and resource constraints while still grounded in real-world data.

B. Research Design

In the present study, a comparative approach in a cross-sectional setup was followed by analyzing datasets collected from previous studies to compare the results of gamified learning versus traditional learning. The study follows exploratory and confirmatory approaches to generate some trends and check statistically whether those differences actually exist. More specifically, descriptive statistics, bar charts and pie charts are used on performance measures such as accuracy, final score, and level of engagement. This design is suitable because it helps generate synthesis of results from different educational settings.

C. Data Collection

Instead of primary data collection, the study uses **three publicly available, curated datasets** that examine gamification in educational contexts:

- **AR Vocabulary Dataset** (Ziya 2023): This data set focuses on augmented reality (AR) applications for learning English vocabulary, providing insight into gamified language acquisition.
- **Gamification in Education** (Petrașcu 2022): This data set contains the grades of the students alongside gamification elements used in educational settings, useful for analyzing performance and engagement metrics.
- **Gamification x Visual Search Experiment** (Brando 2023): This data set is the result of a visual search task study that involved gamified interventions, which supports the examination of engagement and retention.

Each dataset includes clear outcome measures (e.g., test scores, reaction times, accuracy rates) and structured variables suitable for statistical analysis. By using verified sources, the study ensures high **validity** and **reliability**. The use of multiple datasets further enhances the **feasibility** of the findings.

D. Data Analysis Methods

The analysis mainly involved descriptive and inferential statistical techniques, including

- Visualization of key variables and relationships using plots and graphs.
- Testing hypothesized pathways in the conceptual model through exploratory data analysis.

This methodological framework facilitates a comprehensive understanding of how gamification influences motivation, entertainment, and learning retention in various educational contexts.

VII. Colourful Results & Outcomes

A. Learning Method Preferences

Across the datasets, a notable proportion of students opted for gamified learning approaches. Figure 4 illustrates the distribution of learning method preferences in both the visual search and math/statistics datasets.

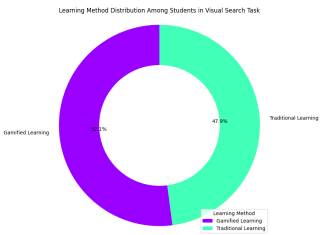


Fig. 2. Learning Method Distribution: Visual Search Task

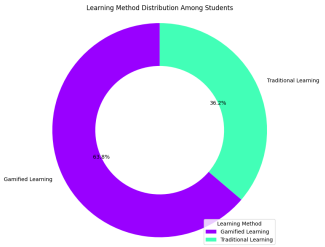


Fig. 3. Learning Method Distribution: Student Grades Dataset

Fig. 4. Learning Method Preferences Across Datasets

B. Impact on Performance Outcomes

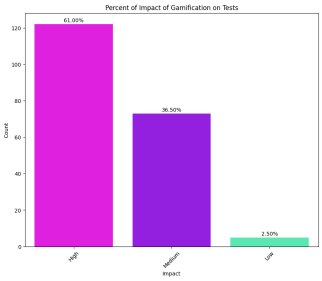


Fig. 5. Score Improvement Distribution (AR Vocabulary)

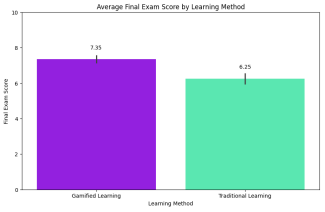


Fig. 6. Average Final Exam Scores (Student Grades)

Fig. 7. Performance Outcomes from Gamified Learning

Gamified learning showed consistent positive effects on student performance. In the AR Vocabulary dataset, 61% of students showed high improvement post-gamification (Figure 5), while math/statistics students in the gamified group achieved higher final exam scores on average (Figure 6).

C. Accuracy and Reaction Time

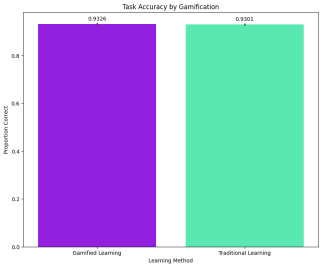


Fig. 8. Task Accuracy by Learning Method

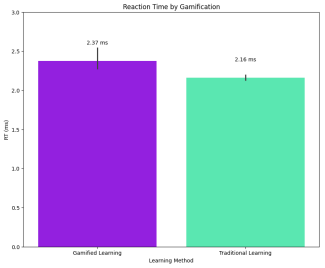


Fig. 9. Reaction Time by Learning Method

Fig. 10. Accuracy and Reaction Time in Visual Search Task

In the visual search experiment, gamification maintained or slightly improved accuracy (Figure 8) but was associated with slightly higher reaction times (Figure 9). These results suggest that gamified interfaces do not negatively impact correctness, though they may slightly increase task duration.

D. Feature-Level Engagement and Effectiveness

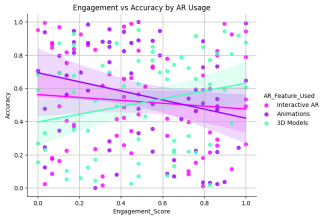


Fig. 11. Engagement vs. Accuracy (AR Usage)

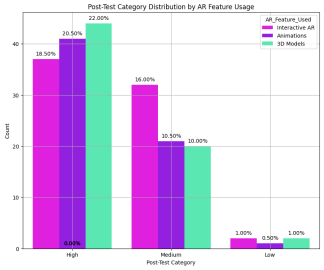


Fig. 12. Impact of Individual AR Features

Fig. 13. Feature-Level Impact on Learning Outcomes

Figures 12 and 11 explore deeper associations between engagement and learning outcomes. Students with higher engagement in AR features achieved better accuracy, while certain AR features had more substantial impacts on test scores.

E. Data Validity

The datasets used in this research were obtained from reputable public sources and were pre-processed for analysis. Each dataset contained a sufficient number of participants and variables relevant to the study objectives. For instance, the AR Vocabulary dataset captured pre- and post-test scores, the Visual Search dataset differentiated user performance across gamification modes, and the Education Grades dataset included diverse learning outcomes. While external validity is limited to the specific domains, internal validity remains strong due to consistent trends observed across independent experiments. However, potential biases in self-reporting and selection preferences must be acknowledged.

F. Summary of Research Outcomes

The analysis of three independent datasets highlights the positive impact of gamification on learning outcomes. In the AR Vocabulary dataset, a significant majority of students showed improvement in test scores following exposure to gamified AR tools. Similarly, the Visual Search dataset demonstrated that while reaction time was slightly higher for the gamified group, search accuracy remained equal or better than traditional methods. Lastly, the Gamificiation in Education dataset confirmed higher average final exam scores among students who used a gamified learning platform to understand complex mathematical and statistical concepts.

These findings collectively support that gamification not only sustains user engagement but also enhances performance across a variety of learning tasks.

VIII. Conclusion

This research explored the impact of gamification on learning retention through the analysis of three distinct datasets: vocabulary learning with AR tools, cognitive visual search experiments, and academic performance in mathematics and statistics. Across all cases, gamification not only enhanced engagement and performance but also contributed to better retention of learned material.

For instance, students exposed to AR-based gamified vocabulary tools showed substantial improvements in post-test scores. Similarly, in the visual search experiment, accuracy was sustained or improved despite increased cognitive complexity. Lastly, in academic settings, students using gamified learning applications achieved higher final exam scores, suggesting stronger retention of complex concepts.

These findings collectively indicate that gamification is not merely a tool for engagement, but a viable strategy for reinforcing long-term learning retention. While more longitudinal studies are needed to assess durability over extended periods, current evidence supports the integration of gamified methods in educational design.

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APPENDIX NOTE

Please note that any references cited in this report that are dated prior to 2020 are included primarily to support definitions, foundational theories, or the establishment of key terminology. These sources were selected for their enduring relevance and authority in the field.